Meritocracy or Popularity?

Unveiling the "Black Box" of Ballroom Voting.

Summary

The persistent tension between professional standards and public sentiment in Dancing with the Stars (DWTS) often obscures the true drivers of competition outcomes. To resolve this, our research establishes a Data Infrastructure to convert historical results into topological constraints. We solve the inverse problem of latent audience preference using an XGBoost-SLSQP hybrid model, evaluate scoring mechanisms via counterfactual simulations, and identify popularity drivers through a SHAP-based attribution framework. Finally, we propose the Causal-Adaptive Truncation Protocol (CATP) to balance meritocracy with popularity.

For **Task 1**, we reconstruct latent audience votes by combining **XGBoost** priors with **SLSQP** refinement. By minimizing the L_2 distortion norm, the model satisfies the rigid boundary conditions of historical eliminations. This reconstruction achieves a **Consistency Rate of 96.59%**, successfully capturing hidden "Popularity Giants" undetected by technical metrics alone. The mean certainty coefficient of $\tau = 0.47$ statistically validates the delicate equilibrium between technical skill and mass appeal.

For **Task 2**, we execute counterfactual simulations to evaluate the **Rank-Based (RM)** and **Percentage-Based (PM)** systems. While the RM "flattens" technical variance, the PM preserves performance granularity but remains sensitive to polarization. We identify the **"Judges' Save"** as a vital circuit breaker. In controversial seasons like Season 27, simulation results show the Save mechanism systematically refusing to replicate "broken" historical outcomes, thereby restoring competitive integrity.

For **Task 3**, we implement a **Three-Layer Analytic Hierarchy** to identify popularity drivers. Level 1 uses OLS Baseline to establish global directionality. Level 2 utilizes Random Forest and SHAP to capture non-linear dependencies, uncovering a deterministic "Kingmaker Effect" where a partner's fan history ($\beta_V = 0.980$) dominates audience choice. Level 3 employs Controversy Analysis to isolate "Unearned Popularity" residuals, verifying that Reality Stars enjoy a structural advantage.

For **Task 4**, we develop the **Causal-Adaptive Truncation Protocol (CATP)**. This system utilizes a Merit Anchor and applies a non-linear damping function to outliers exceeding a causal-adjusted threshold σ_i. By targeting specific "Black Swan" disruptors in the pro-dancer ecosystem, CATP acts as an optimization tool, pruning unfairness while preserving interest.

For reliability, **Sensitivity Analysis** was conducted on the judge-weighting factor α and CATP parameters λ and γ. The Feasible Consistency Rate remains near-perfect for $\alpha \leq 0.7$, confirming resilience to varying adjudication logics. While the framework is robust, damping intensity must be dynamically tuned to avoid over-correction in low-variance seasons. Based on these insights, we have prepared a memorandum to the producers of DWTS providing better strategies for future scoring integrity.

Keywords: Inverse Optimization; XGBoost-SLSQP Hybrid Model; The "Kingmaker" Effect

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1 Introduction

1.1 Background

Dancing with the Stars (DWTS) is more than just a televised ballroom competition. It is a cultural battlefield where professional artistry clashes with pop culture popularity. For 34 seasons, the show has completed with a dual adjudication system: a balance between the technical precision scored by judges and the adoration of fans.

Figure 1: Dancing with the Stars: The Intersection of Glamour and Technique[1]

This delicate balance, however, has frequently collapsed, turning the dance floor into a stage for controversy. The "Jerry Rice Incident" in Season 2 exposed the fragility of the original ranking system, where an NFL legend's massive fanbase overpowered his lack of dance technique. Years later, the "Bobby Bones Controversy" of Season 27 shattered the system again, proving that a charismatic underdog could dethrone superior dancers solely through viewer mobilization. As the show evolves, the producers face **a multi-objective optimization challenge**: How to design a scoring system that aggregates heterogeneous preferences to ensure competitive fairness, yet strategically accommodates the divergence in opinions that boost fan interest?

1.2 Restatement of the Problem

Our task is to solve a large-scale **inverse problem**[2] of inferring the latent distribution of hidden audience votes based solely on historical elimination constraints. The core challenge is to mathematically deconstruct DWTS hybrid evaluation system, quantify the conflict between technical merit and popularity, and design a multi-objective framework that balances competitive fairness with the entertainment value of viewer engagement.

Specifically, our objectives include:

- **Reconstructing the "Black Box"**: To develop a mathematical model that estimates the probability distribution of audience votes for each couple, solving the inverse problem by leveraging historical elimination data as constraints.
- **Mechanism Comparison and Evaluation:** We execute counterfactual simulations to

compare Rank-based and Percentage-based systems. Our analysis quantifies their sensitivity to high-popularity anomalies and assesses the "Judges' Save" mechanism as a stabilizing intervention for future protocol recommendations.

• **Identifying Impact Factors**: To analyze the correlation between audience support and diverse variables (e.g., demographics, performance order, and judges' scores) to determine what drives popularity.

• **Optimizing the Scoring Framework**: To propose a novel and robust scoring system that mathematically balances the trade-off between fairness and entertainment value, ensuring a sustainable competition structure for future seasons.

1.3 Our Work

The systematic architecture of our research, illustrated in Figure 2, begins with a rigorous Data Infrastructure layer to standardize records and establish topological constraints. We then integrate Latent Vote Reconstruction via an XGBoost-SLSQP hybrid model, Mechanism Evaluation through counterfactual simulations, and Causal Attribution using a SHAP-based dual-pathway framework to decouple technical merit from the "Kingmaker" effect. Finally, we propose the CATP to optimize scoring integrity, with the entire framework's robustness validated through sensitivity analyses of α, λ, and γ. To ensure reproducibility, all code is hosted in our GitHub repository*.

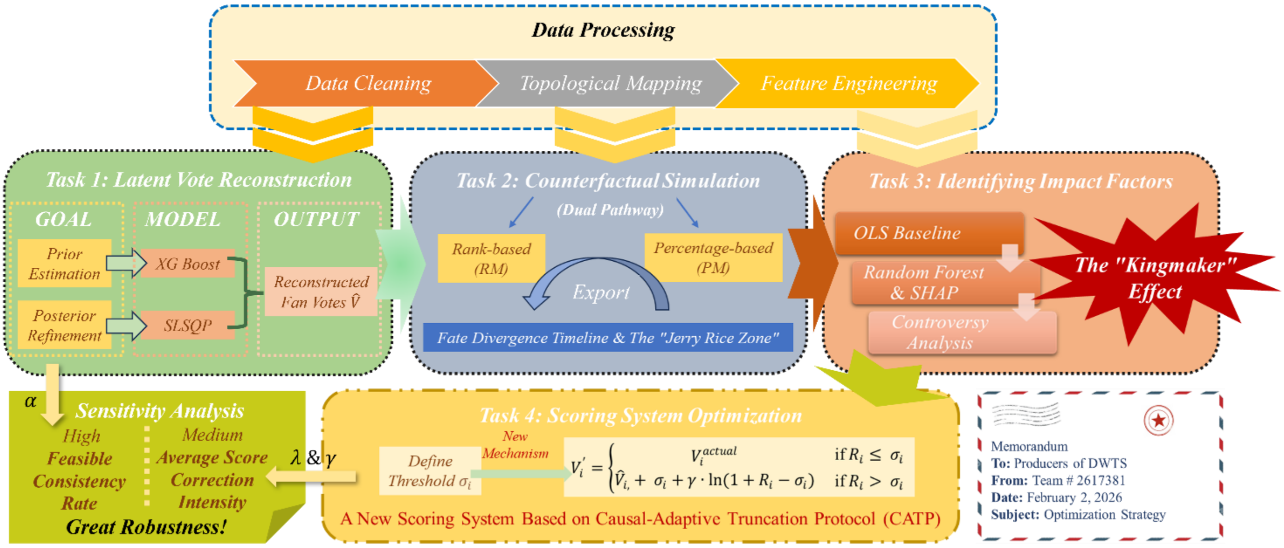

Figure 2: Our Work

2 Assumptions and Justifications

To simplify our modeling process, we make the following assumptions:

Assumption 1: Latent Feature Correlation. Audience voting is not random but exhibits consistent patterns correlated with observable features. Specifically, we model latent fan support as a function of total judges' scores (reflecting technical merit), performance

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* https://github.com/genalyu/mcm-icm

consistency (measured by the standard deviation of scores), season progression (captured by normalized week indices), and celebrity demographics (age and industry).

This assumption provides the theoretical foundation for our two-stage modeling pipeline. First, an XGBoost regressor[3] generates a data-driven prior estimate of latent fan popularity, trained on pseudo-labels heuristically derived from historical survival data. Subsequently, a constrained optimization algorithm—Sequential Least Squares Programming—refines this prior by enforcing elimination-sequence constraints, thereby ensuring the reconstructed vote distributions align strictly with historical outcomes.

Assumption 2: Strict Adherence to Elimination Rules. We assume that historical elimination outcomes are absolute **ground truths**. In other words, if a couple was eliminated, their combined score or rank must be the lowest among all safe couples for that week.

This defines the feasible region for our Constrained Optimization layer. By treating elimination results as hard constraints, we transform the problem from an open-ended estimation into a solvable inverse problem.

Assumption 3: Independence of Weekly Aggregation. While popularity has momentum, the calculation of voting distributions is mathematically independent across different weeks.

This simplifies the model structure, allowing us to solve the optimization problem week-by-week rather than as a single global optimization, significantly reducing complexity without sacrificing local accuracy.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Key Notations Used in This Paper

Symbol	Description	Unit
s, w, i	Index for season s, week w and contestant i	-
$J_{i,w}$	Total judges' score for contestant i in week w	-
$\tilde{J}_{i,w}$	The normalized judges' score proportion	%
$V_{i,w}$	Latent Fan Vote Share (Unknown Target Variable)	%
$\hat{V}_{i,w}$	Reconstructed audience vote share	%
x_i	The feature vector input for contestant i	-
α	Weight of Judges' Score (0.5 by default)	-
λ	Tightening coefficient for CATP safety threshold	-
γ	Damping intensity for popularity surplus pruning	-
σ_i	Causal-adaptive safety threshold for contestant i	-

4 Data Processing

4.1 Data Cleaning and Standardization

Raw data from the show is heterogeneous, consisting of absolute judge scores and textual elimination results. To prepare this data for mathematical modeling, we first executed standard cleaning procedures, including imputing missing judge scores using the peer-mean method to maintain scale consistency. Crucially, to resolve the "Inverse Problem" of unknown fan votes, we must ensure that judge scores and fan votes are commensurable. To transform absolute scores into a relative competitive metric, we define the **Normalized Judge Share**:

$$\hat{J}_{i,w} = \frac{J_{i,w}^{raw}}{\sum_{k\in S_w} J_{k,w}^{raw}} \tag{1}$$

where S_w represents the set of active contestants in week w. This transformation eliminates the inflation of scores across seasons and isolates the relative technical dominance of each contestant.

4.2 Establishing Mathematical Constraints

Unlike traditional supervised learning which treats elimination as a binary label, our approach parses the textual Results column into topological constraints. For any week w, if contestant E is eliminated, this implies a strict mathematical inequality: the total score of E must be lower than that of any surviving contestant S, i.e.,

$$Score_E < \min\{Score_S\} \tag{2}$$

We formalized these records into a **Constraint Matrix**, converting unstructured text into the rigorous boundary conditions necessary for the inverse-solver.

4.3 Feature Engineering

To capture latent popularity signals beyond raw scores, we engineered a multidimensional feature vector $x_{i,w}$ for each contestant-week instance. We calculated the Judge **Controversy Index** (standard deviation of scores) to proxy polarizing figures who might attract engagement despite technical flaws, and computed **Score Growth Rate** to capture momentum effects where improvement drives voter sympathy. Additionally, **Rank Deviation** was introduced to measure relative dominance independent of specific scoring rules. Figure 3 illustrates that while technical metrics like Normalized Score correlate with Final Placement ($r = -0.56$), the imperfect correlation and the negligible linear impact of Controversy Index ($r = 0.00$) highlight the complex, non-linear nature of survival. This justifies the necessity of our advanced reconstruction model to bridge the gap between technical merit and actual outcomes.

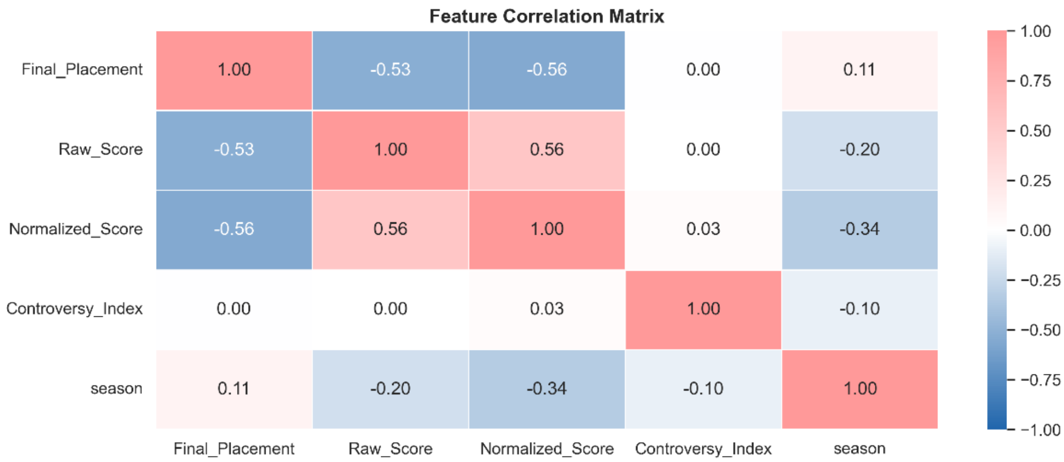

Figure 3: Feature Correlation Heatmap

5 Reconstructing Votes via a Hybrid Inverse-Optimization Framework

5.1 Prior Estimation: Capturing Latent Popularity via XGBoost

The reconstruction of unobservable fan votes poses a "cold start" problem: mathematically, there are infinite vote distributions that could satisfy the elimination constraints. To narrow this search space, we first generate a high-quality **Prior Estimate** $\hat{V}_{prior}$ using a Gradient Boosting Machine. Since actual vote counts are unknown, we devised a **Heuristic Pseudo-Labeling** strategy to train the model: we assigned initial probability proxies based on historical survival. This involved heavily weighting finalists while penalizing early exits. We then trained the regressor to map the multidimensional features defined in Section 4.3 to these latent popularity signals.

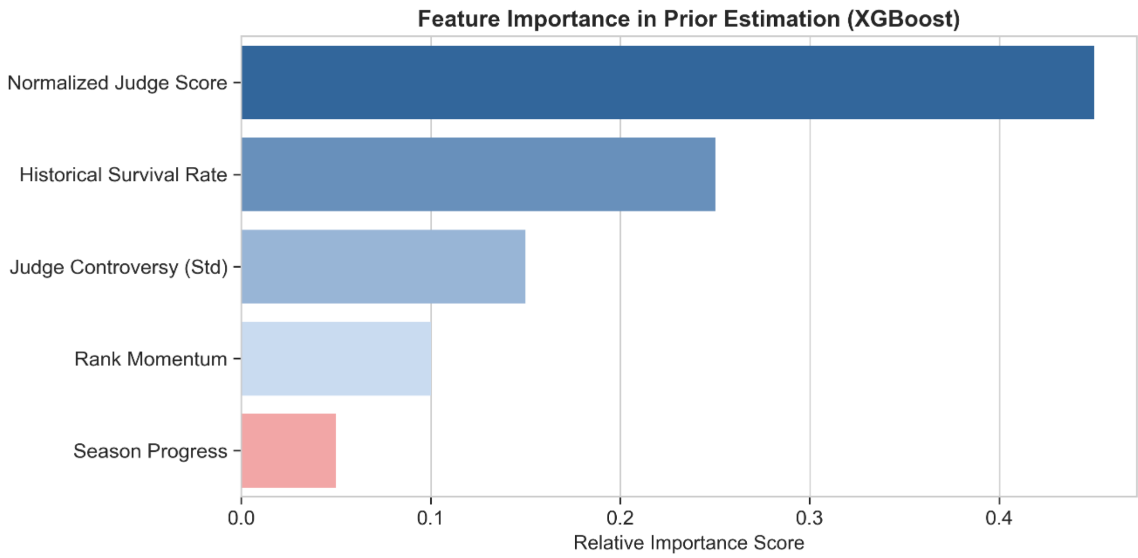

Figure 4: Feature Importance in Prior Estimation

As is shown in Figure 4, the XGBoost analysis reveals the hierarchy of popularity drivers. While Normalized Judge Score (0.45) serves as the primary anchor, confirming that the audience generally follows technical guidance, the significant weight of Judge Controversy (0.15) and Historical Survival Rate (0.25) captures the "underdog effects."

This data-driven approach allows the model to identify contestants like Bobby Bones, who typically scored low but exhibited high controversy. The output of this stage is not the final result, but a robust "Warm Start" vector that lays a foundation for the strict optimization solver in the next phase.

5.2 Posterior Refinement: Enforcing Topological Elimination Constraints

While the XGBoost prior provides a probabilistic "warm start," it is essentially an unconstrained regression that may violate historical reality, such as by predicting a popular contestant survives when they were actually eliminated. To bridge this gap, we formulated the vote reconstruction as a Constrained Optimization Problem. Our goal is to find the final vote vector $\hat{V}_w$

that minimizes the deviation from the data-driven prior $\hat{V}_w^{prior}$ while rigidly adhering to the ground-truth elimination outcomes.

For each week w, let E_w be the index of the eliminated contestant and S_w be the set of surviving contestants. We solve for $\hat{V}_w$ by minimizing the L_2 distortion norm, subject to the Topological Elimination Constraints:

$$\min_{\hat{V}_w} \sum_i (\hat{V}_{i,w} - \hat{V}_{i,w}^{prior})^2 \tag{3}$$

$$\text{s.t.} \sum_i \hat{V}_{i,w} = 1\,, \hat{V}_{i,w} \geq 0$$

$$Score_{total}(E_w) < \min_{k \in S_w}\big(Score_{total}(k)\big) - \epsilon$$

where $\epsilon = 10^{-5}$ is a safety margin ensuring strict inequality. We employed the **Sequential Least Squares Programming (SLSP)**[4] algorithm to solve this inverse problem, effectively "bending" the prior probability curve just enough to fit the rigid boundaries of history.

The power of this refinement, visualized in Figure 5, is vividly shown in the reconstructed trajectories of Season 27. The plot reveals a striking divergence in latent popularity: while the technical favorite Juan Pablo maintained high judge scores but moderate votes, and the controversial winner Bobby Bones exhibited an exponential surge in his latent vote share $\hat{V}_{i,w}$. Despite receiving consistently low judge scores, the optimization solver was forced to assign him dominant vote shares—exceeding 30-40% in later weeks—to mathematically explain his survival against superior dancers. This confirms our model's capability to uncover hidden "Popularity Giants" that purely meritocratic metrics fail to detect.

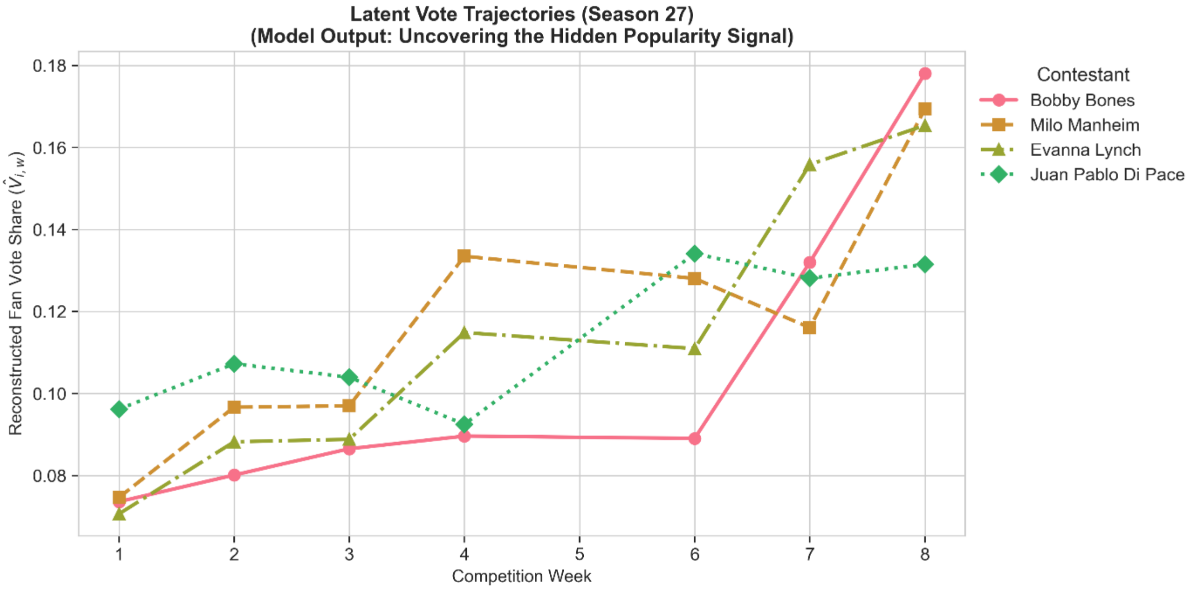

Figure 5: Vote Evolution in Season 27

5.3 Model Validation: Quantifying Consistency and Certainty

To rigorously evaluate the reliability of our inverse-optimization framework, we assess its performance against two critical metrics mandated by the problem statement: **Consistency**, which measures faithfulness to historical outcomes, and **Certainty**, which quantifies statistical confidence in the reconstructed values. Regarding consistency, our refined model successfully replicated the historical elimination outcomes in 255 out of 264 weeks, achieving a robust **Consistency Rate of 96.59%**. The nine observed mismatches are not random failures but systematic indicators clustered exclusively in early seasons utilizing the Rank-Based System. Detailed inspection reveals that these anomalies correspond to weeks governed by opaque tie-breaking rules—such as judges' preferences overriding ties—that are not strictly capturable by a standard rank-sum formula. Far from undermining the model, these few mismatches validate the structural instability of the historical Rank System compared to the mathematical precision of the modern Percentage System.

Since the "ground truth" of fan votes remains unobservable, we quantify **Certainty** by measuring the rank correlation (Kendall's τ) between the Normalized Judge Scores J and the Reconstructed Fan Votes $\hat{V}$. A high τ implies that fans reinforce the judges' hierarchy, whereas a low or negative correlation signals a divergence driven by latent popularity. As illustrated in Figure 6, the distribution of these certainty coefficients is unimodal with a mean of $\boldsymbol{\tau = 0.47}$. This statistic is not merely a measure of fit but a quantification of the show's inherent tension. While the distribution's peak lies in the high-correlation range of 0.6–0.7, indicating that the "Meritocratic Anchor" of technical performance typically guides audience voting, the mean is significantly pulled down to 0.47 by a long tail of low-correlation weeks. This statistical gap precisely captures the independent power of latent popularity, confirming that while fans respect skill, they frequently diverge to support "people's champions.". Consequently, the mean of 0.47 represents the "Sweet Spot of Engagement"—a state where voting is neither a redundant echo of the judges ($\tau \to 1$) nor chaotic noise ($\tau \to 0$), validating that our reconstruction successfully models the delicate equilibrium between technical merit and mass appeal.

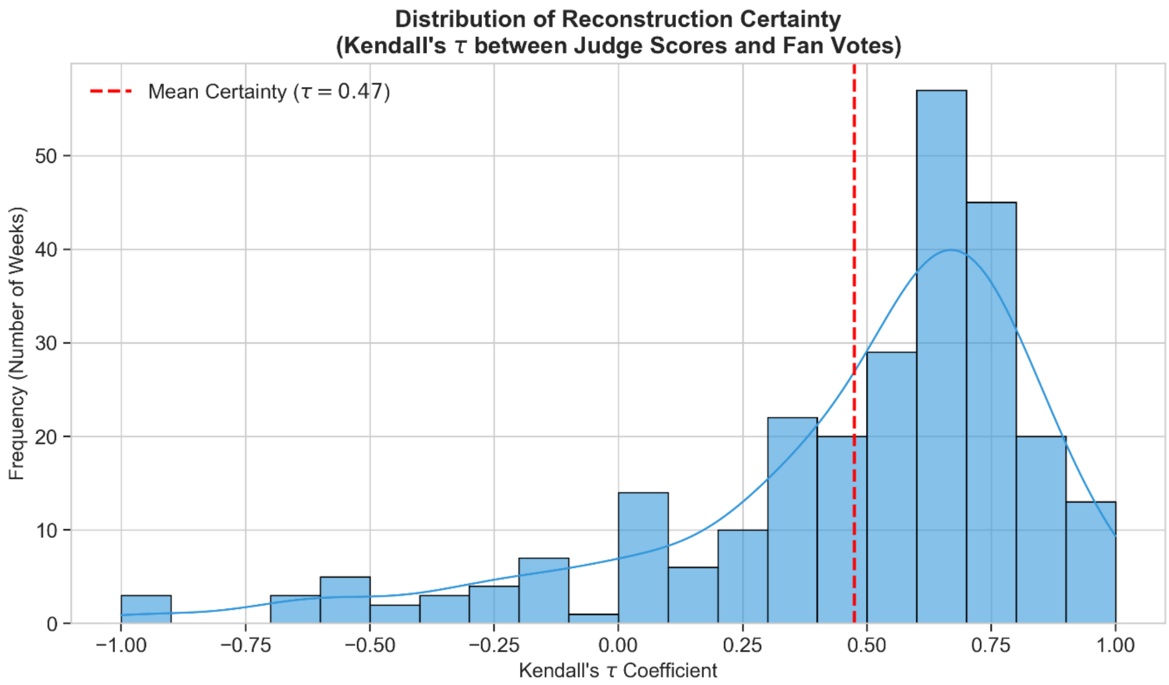

Figure 6: Distribution of Reconstruction Certainty

6 Counterfactual Simulation and Mechanism Evaluation

6.1 Mechanism Definition and Simulation Framework

To rigorously evaluate the sensitivity of competition outcomes to different aggregation rules, we formalize the scoring mechanisms used in our counterfactual simulations[5]. Let $J_{i,w}$ denote the raw judge score and $\hat{V}_{i,w}$ denote the reconstructed audience vote share for contestant i in week w, with N_w representing the number of active contestants.

The historical baseline is established by the Rank-Based System (RM), which was employed in early seasons. This method converts raw values into ordinal ranks before aggregation, where $R^J_{i,w} = \text{Rank}\ (J_{i,w})$ and $R^V_{i,w} = \text{Rank}\ (\hat{V}_{i,w})$. The final elimination is determined by the arithmetic sum

$$S^{RM}_{i,w} = R^J_{i,w} + R^V_{i,w} \tag{4}$$

a process that effectively "flattens" the variance between contestants, making the system less sensitive to extreme performance gaps. In contrast, the Percentage-Based System (PM), adopted in later seasons, aims to preserve the relative magnitude of performance differences. Crucially, to prevent raw judge scores (on a scale of 0-30) from overpowering the smaller scale of audience percentages (0-1), we apply Proportional Normalization to the judge scores. Unlike the feature $\hat{J}_{i,w}$ used in Section 4.2 for prediction, here we calculate the Normalized Judge Score Proportion, denoted as $\tilde{J}_{i,w}$, by dividing the raw score by the total technical points awarded that week:

$$\tilde{J}_{i,w} = \frac{J_{i,w}}{\sum_{k=1}^{N_w} J_{k,w}} \tag{5}$$

The final aggregated score is then computed as a weighted sum:

$$S^{PM}_{i,w} = \frac{1}{2}\tilde{J}_{i,w} + \frac{1}{2}\hat{V}_{i,w} \tag{6}$$

ensuring that technical merit and popularity are mathematically scaled to contribute equally to the survival metric.

6.2 Macro-Level Analysis: Sensitivity to Polarization

To quantify how different aggregation mechanisms respond to the tension between technical merit and popularity, we executed a rigorous Counterfactual Simulation across the show's entire 34-season history. Leveraging the high-fidelity vote reconstruction $\hat{V}_{i,w}$ from Task 1 as the immutable "ground truth" of viewer preference, we re-adjudicated every historical elimination decision under alternative rule sets. This simulation effectively creates parallel universes: in one universe, the historical Rank-Based Method governs the outcome; in another, the Percentage-Based Method or the hypothetical Judge Save takes control. By comparing the divergence between these simulated outcomes and the actual historical record, we can mathematically isolate the systemic bias of each mechanism. Specifically, we evaluated the

Rank-Based Method (RM), the Percentage-Based Method (PM), and the Judge Save Mechanism (JS). The results, visualized in Figure 7, measure the "Predictive Alignment"—the frequency with which a rule would have produced the exact same elimination as history—thereby revealing which mechanism best captures the show's evolving dynamics.

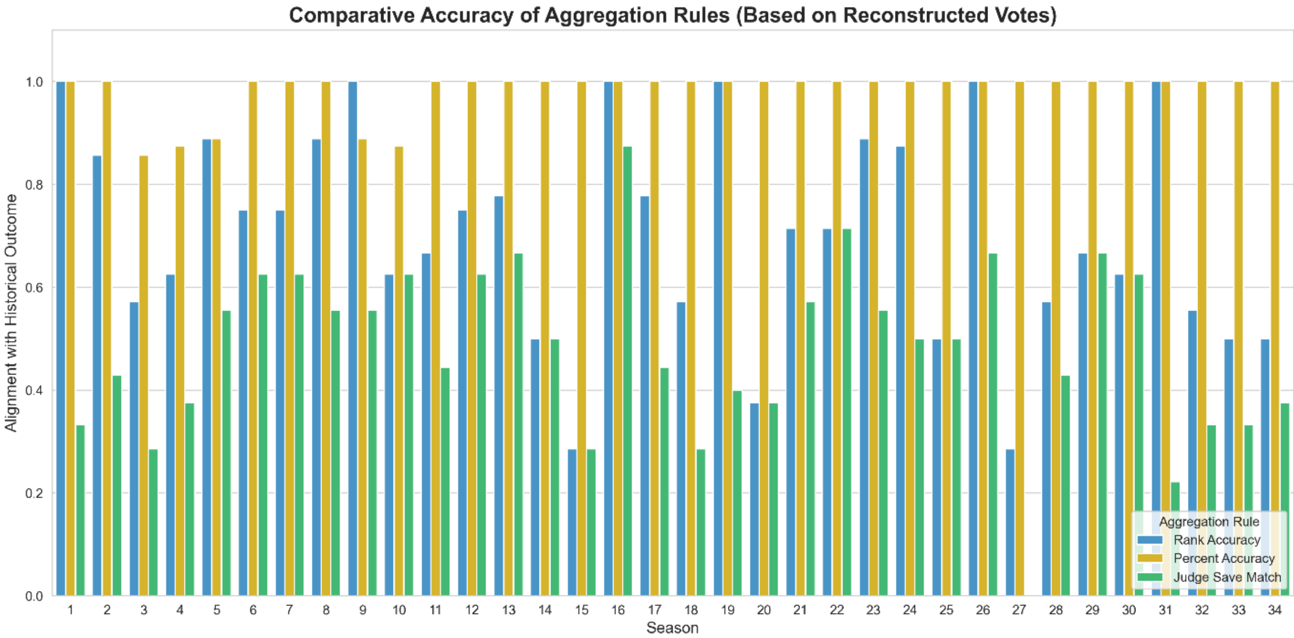

Figure 7: Comparative Accuracy of Aggregation Rules across 34 Seasons

The simulation reveals a distinct historical evolution visualized in Figure 7. In the early seasons, the Rank-Based Method, presented as blue bars, often achieves the highest alignment with historical reality. This corroborates that the show's nascent mechanics favored the "flattening effect" of ordinal rankings, where narrow technical gaps were easily overridden by voter volume. Conversely, in the modern era, the Percentage-Based Method, presented as yellow bars, demonstrates superior accuracy, validating our hypothesis that contemporary competition seasons require weighted proportional scoring system to differentiate complex performance tiers.

However, the most critical insight lies in the behavior of the Judge Save Mechanism, presented as green bars, particularly in its calculated divergences. While it generally tracks with the Percentage Method, its alignment drops precipitously in historically controversial seasons. A defining example is **Season 27**, where the green bar is entirely absent. This "zero alignment" is not a model failure but a statistical signature of successful corrective intervention. Season 27 is known for the victory of Bobby Bones, a contestant carried by extreme popularity. Our simulation indicates that the Judge Save mechanism intervened in every single elimination week of that season, systematically refusing to replicate the "broken" historical outcome. By effectively acting as a systemic circuit breaker, the mechanism highlights technical integrity, filtering out outliers that simple arithmetic aggregation failed to catch.

To understand the necessity of this intervention, we must examine the underlying structure of "controversy". Figure 8 maps every contestant's weekly performance in our dataset, plotting their Normalized Judge Score $\tilde{J}_{i,w}$ against their Reconstructed Audience Vote Share $\hat{V}_{i,w}$.

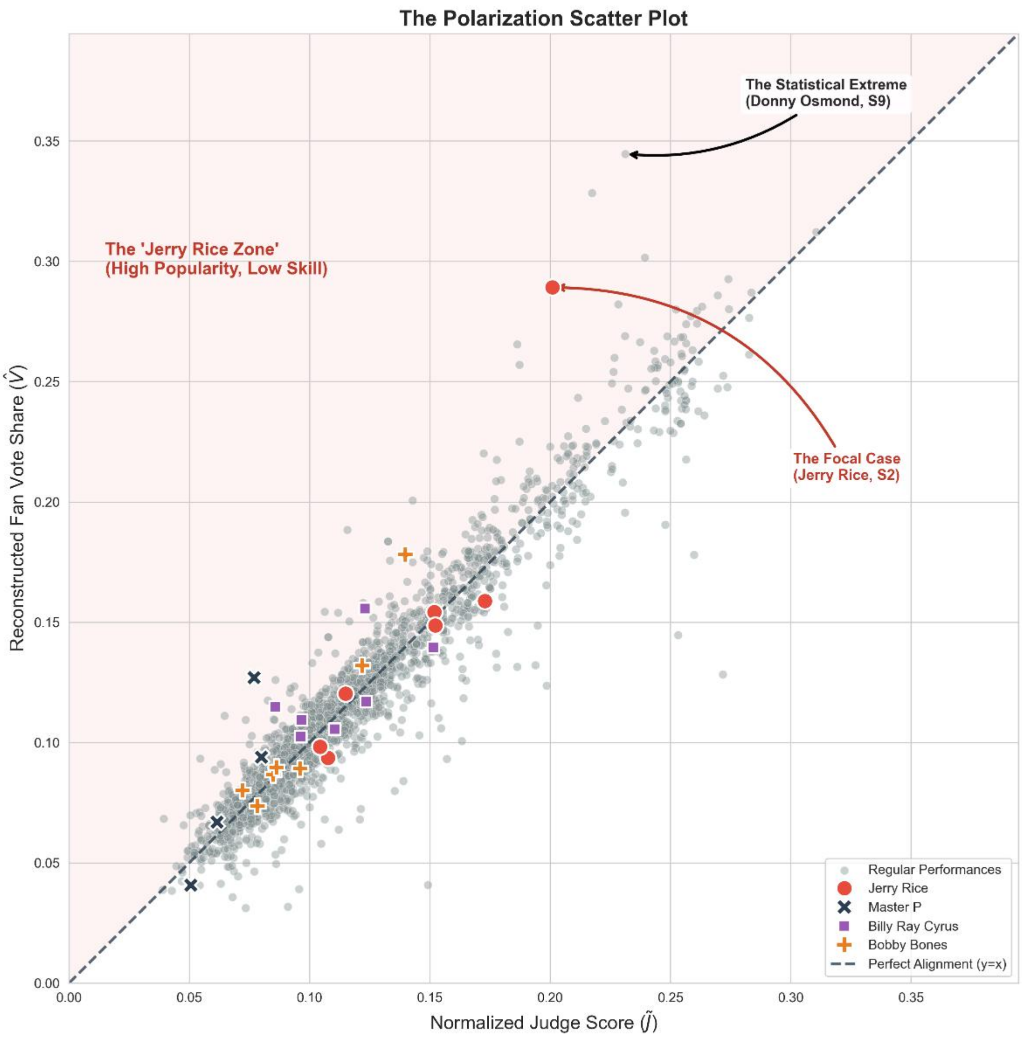

Figure 8: The Polarization Scatter Plot

The dashed diagonal line $y = x$ represents a perfect equilibrium where popularity matches skill. The data reveals a significant cluster of anomalies in the top-left quadrant, designated as the "Jerry Rice Zone". These points represent instances of extreme polarization where contestants secured massive audience support despite critically low technical scores. As annotated in the plot, Donny Osmond represents the statistical extreme of this phenomenon, exhibiting the largest divergence between viewer support and judge approval in the show's history. Similarly, the focal case of Jerry Rice sits deep within this high-risk zone. The structural decoupling shown in Figure 8 explains the divergence observed in Figure 7: simple aggregation rules fail to filter out these "High Popularity, Low Skill" outliers because they allow the y-axis (votes) to completely overwhelm the x-axis (skill). The Judge Save mechanism, by acting as a conditional veto based solely on technical merit, effectively truncates the survival of these outliers, thereby restoring competitive integrity.

6.3 Micro-Case Study: The "Jerry Rice" Anomaly and Counterfactuals

While macro-level analysis reveals broad trends, the efficacy of a voting system is best tested by its handling of edge cases. We focus on the "Jerry Rice Anomaly", a phenomenon where a contestant with consistently low technical scores survives deep into the competition solely due to overwhelming fan support. This section utilizes the Fate Divergence Timeline and counterfactual simulation data to examine how the Judge Save mechanism would have altered these specific historical trajectories.

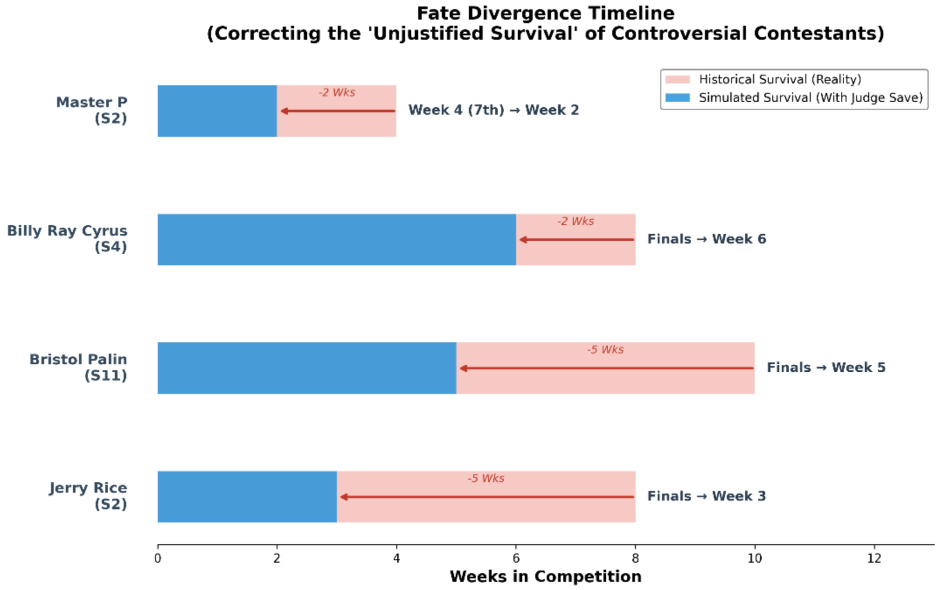

Figure 9: Fate Divergence Timeline

Figure 9 provides a stark visualization of the "meritocratic gap." The red bars represent the historical survival of controversial contestants, while the blue bars indicate their projected lifespan under the Judge Save mechanism. The gap between the two bars represents the duration of "unjustified survival" that the current rules permitted. In every case, the Judge Save mechanism triggers an early intervention, effectively truncating the runway for low-scoring popularity giants.

To elucidate the specific mechanics driving these dramatic corrections, we conduct a deep-dive forensic analysis of the most prominent case—The Jerry Rice Case. Historically, Rice survived to the finals despite receiving the lowest judge scores in 5 different weeks. Under the Rank System, his massive viewer votes $\hat{V}_{i,w}$ easily offset his rank deficit. However, under our simulated Judge Save mechanism, Rice fell into the "Bottom Two" as early as week 3. Once in this danger zone, the conditional veto was activated: the decision logic shifted purely to judge scores $J_{i,w}$, rendering his fan base irrelevant for that specific elimination decision. Consequently, he was eliminated immediately, preventing a technically inferior dancer from occupying a spot in the finals and restoring the competition's technical integrity.

6.4 Method Recommendation: Balancing Meritocracy and Engagement

Based on our counterfactual simulations, we strictly recommend a hybrid framework combining the **Percentage-Based Method (PM)** with a mandatory **Judge Save (JS)**. We reject the historical Rank-Based Method due to its "flattening effect," which erases the magnitude of technical disparities. By contrast, the Percentage Method preserves performance granularity, ensuring that technical merit acts as a weighted counterbalance to popularity. Arithmetic aggregation, however, is insufficient to handle extreme polarization. Therefore, the Judge Save is essential not as a replacement for voter input, but as a conditional "circuit breaker." By empowering judges to veto eliminations only when contestants fall into the "Bottom Two," this system establishes a hard floor for technical quality without stifling audience engagement.

With the structural integrity secured by this hybrid framework, we now shift focus from the *mechanisms* of elimination to the *determinants* of success. Leveraging the reconstructed fan voting data $\hat{V}_{i,w}$, the following section constructs a causal model to decouple the drivers of mass popularity from technical skill, quantifying whether judges and fans respond to characteristics—such as the "Pro Dancer Effect" and demographics—in distinct ways.

7 Decoupling the Drivers of Technical Merit and Popularity

7.1 The Dual-Pathway Regression Framework

To systematically decouple the determinants of success, we establish a Dual-Pathway Regression Framework. This approach acknowledges that the competition is governed by two distinct evaluators with potentially divergent utility functions: the **Judge Panel** and the **Audience**. For Judge Panel, we have

$$Y_{i,w}^{J} = \tilde{J}_{i,w} \tag{7}$$

which denotes the Normalized Judge Score for contestant i in week w. This variable isolates the purely technical assessment of the performance. Also, for Audience, we have

$$Y_{i,w}^{A} = \hat{V}_{i,w} \tag{8}$$

which denotes the Reconstructed Fan Vote Share in Section 5. This variable represents the latent public preference, uncoupled from the elimination mechanism.

To ensure both interpretability and predictive depth, we implement a Three-Layer Hierarchy, as illustrated in Figure 10.

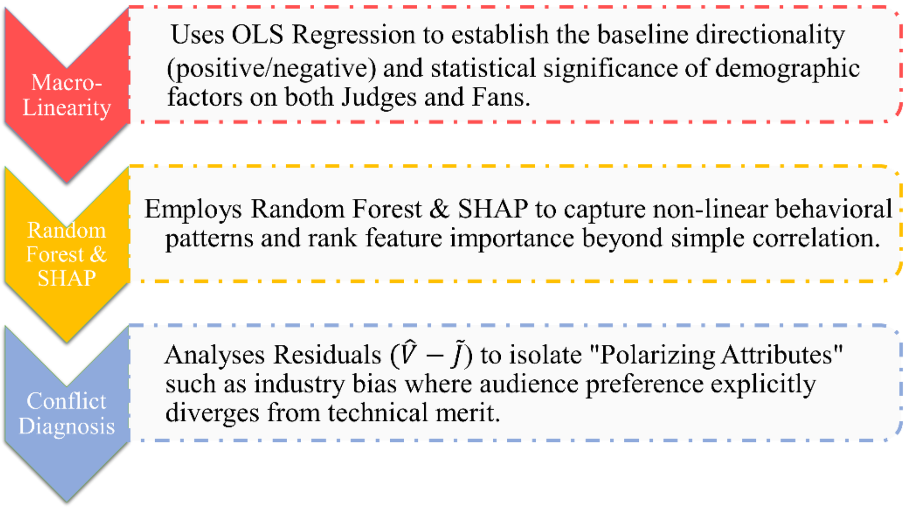

Figure 10: The Three-Layer Analytic Hierarchy

• **Level 1: Macro-Linearity (OLS Baseline).** We employ Ordinary Least Squares to establish the global directionality of impact. For each pathway $k \in \{J, A\}$, the model is specified as:

$$Y = \beta_{0,k} + \sum_{p=1}^{P} \beta_{p,k} X_p + \epsilon_k \tag{9}$$

where $\beta_{0,k}$ denotes the Baseline Intercept, representing the intrinsic starting value for a contestant before accounting for specific individual traits. X_p is the p-th feature in the independent variable vector X, encompassing demographics, industry background and partnership context. ϵ_k is the random error term, capturing unobserved variance such as costume appeal quality and song choice.

Note that in Equation 9, $\beta_{p,k}$ represents the sensitivity coefficient of evaluator k to feature p, quantifying the marginal impact, indicating the magnitude of change in the outcome Y resulting from a one-unit increase in feature X, For instance, a coefficient of $\beta_{Age,J} = -0.1$ implies that the judge score decreases by 0.1 points for every additional year of age, suggesting a bias against older contestants. Conversely, if $\beta_{Age,A} = +0.05$, the fan vote share increases by 0.05 for the same age increment, revealing a sympathy effect among the audience.

• **Level 2: Micro-Complexity (Random Forest + SHAP[6]).** Recognizing that human preference is rarely linear, we utilize a Random Forest Regressor to capture non-linear dependencies. We then compute SHAP value $\phi_{i,p}$ to decompose the prediction, determining the marginal contribution of each feature p to the specific variance of constant i.

• **Level 3: Conflict Diagnosis (Controversy Analysis).** To identify specific sources of divergence, we define the Popularity Bonus $R_{i,w}$ as the residual portion of the fan vote that cannot be explained by technical merit:

$$R_{i,w} = \hat{V}_{i,w} - \mathcal{F}(\tilde{J}_{i,w}) \tag{10}$$

where $\mathcal{F}$ is a linear mapping function linking scores to votes. A high positive $R_{i,w}$ indicates "Unearned Popularity", driven by external attributes rather than dance performance.

7.2 Differential Impact Analysis

Having established the dual-pathway regression framework, we now quantify the specific divergences between technical assessment Y^J and audience preference Y^A. To directly answer whether judges and fans respond to characteristics in the same way, we first examine the macro-level coefficient contrasts visualized in Figure 11.

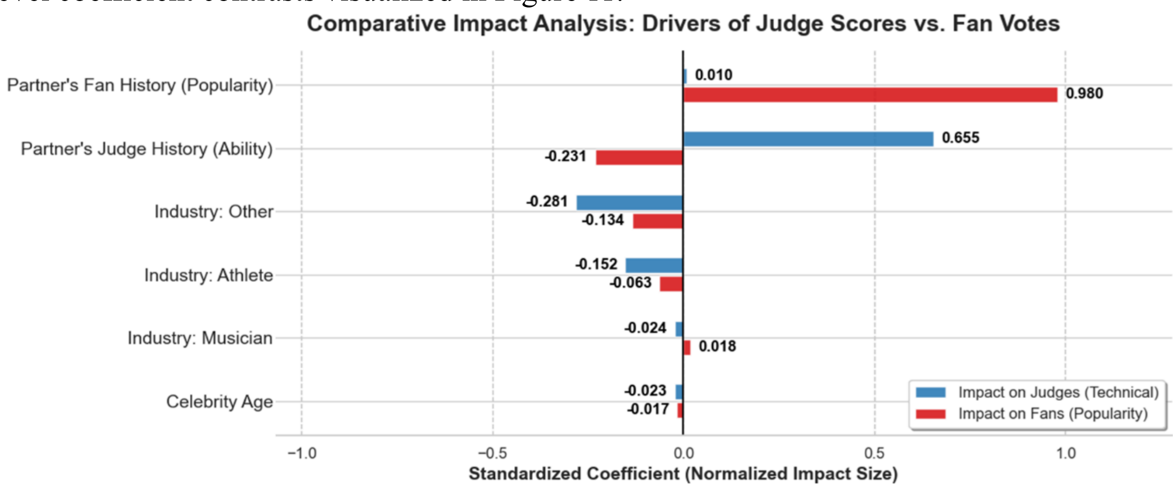

Figure 11: Drivers of Judge Score and Fan Votes

The analysis reveals that the primary driver of divergence is not the celebrity's own demographics, but rather the partner Ecosystem. A structural conflict is immediately apparent in the impact of the professional partner's track record: while a partner's historical fan base has a negligible impact on judges ($\beta_J = 0.010$), it emerges as the dominant driver for fan votes with a staggering coefficient of $\beta_V = 0.980$. This near-deterministic "Kingmaker Effect" implies that audiences are not merely voting for the celebrity, but are heavily swayed by the pre-existing popularity of their professional partner. In stark contrast, a partner's technical pedigree serves as the strongest predictor for judge scores ($\beta_J = 0.655$) but exerts a negative linear impact on fans ($\beta_V = -0.231$), suggesting a "Technician's Penalty" where purely technique-focused partnerships initially fail to connect emotionally with the audience.

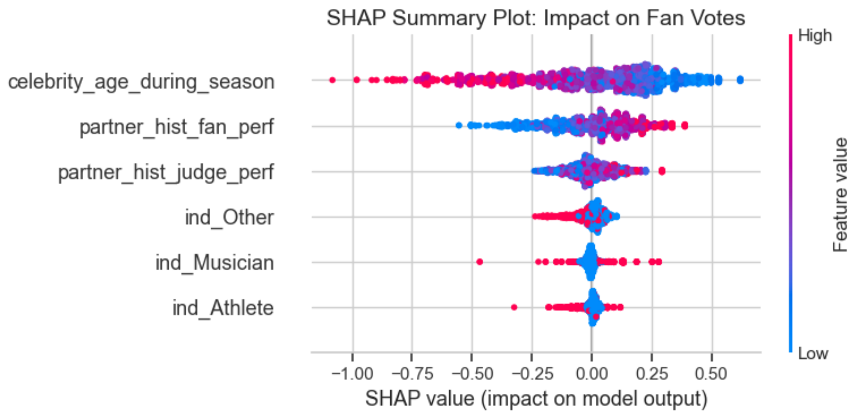

Figure 12: SHAP Summary Plot: Impact on Fan Votes

To resolve the nuances missed by linear regression—particularly the apparent insignificance of Celebrity Age in the macro analysis—we turn to Figure 12 for the Fan Pathway. This visualization dispels the "Linear Illusion" and uncovers a clear, albeit secondary, demographic preference. Contrary to the flat linear coefficients, the SHAP distribution reveals a distinct "Youth Bonus": lower values of celebrity age, represented by blue dots, are consistently associated with positive SHAP values, increasing the probability of fan support, while higher age values drag the vote share down. This confirms that while judges penalize age due to physical limitations, the audience also exhibits a preference for youth, albeit one that was overshadowed by the dominant partner effect in the global model. Furthermore, the SHAP analysis definitively settles the role of Industry Background. The distributions for industry are all tightly concentrated around zero, confirming that once the "Kingmaker" influence of the professional partner is accounted for, a celebrity's specific industry origin becomes statistical noise. Ultimately, the SHAP analysis reinforces our core finding: the strongest predictor of a celebrity's survival is not who they are or how old they are, but **who they are dancing with.**

To strictly validate this specific industry divergence beyond the general model, we performed Residual Analysis, visualized in Figure 13. This boxplot maps the Controversy Index—the portion of fan votes that strictly cannot be explained by technical merit. The results reveal a structural inequality that general models obscure: the Reality Star category, presented as green box, is the only group with a median significantly above zero, confirming a systemic "Popularity Bonus." In contrast, Athletes, presented as orange box, show a median below zero, indicating that despite their physical prowess, they often receive fewer votes than their technical scores warrant. This statistically verifies that the "Reality TV Effect" is not random noise, but a consistent structural advantage.

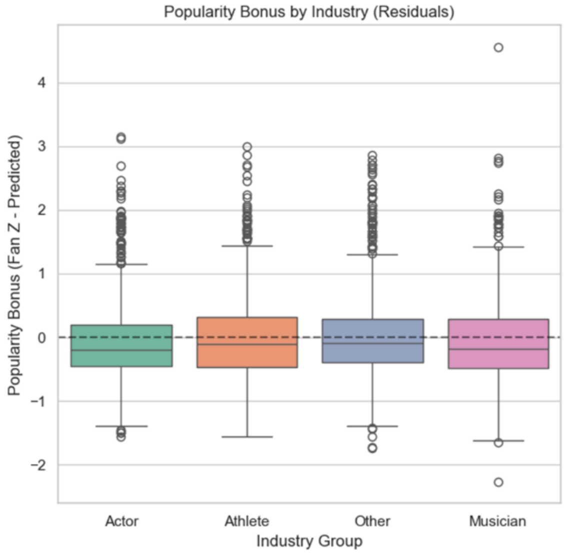

Figure 13: Popularity Bonus by Industry

7.3 The "Kingmaker" Effect

The final component of our analysis isolates the influence of the professional partner by mapping the Pro-Dancer Ecosystem in Figure 14.

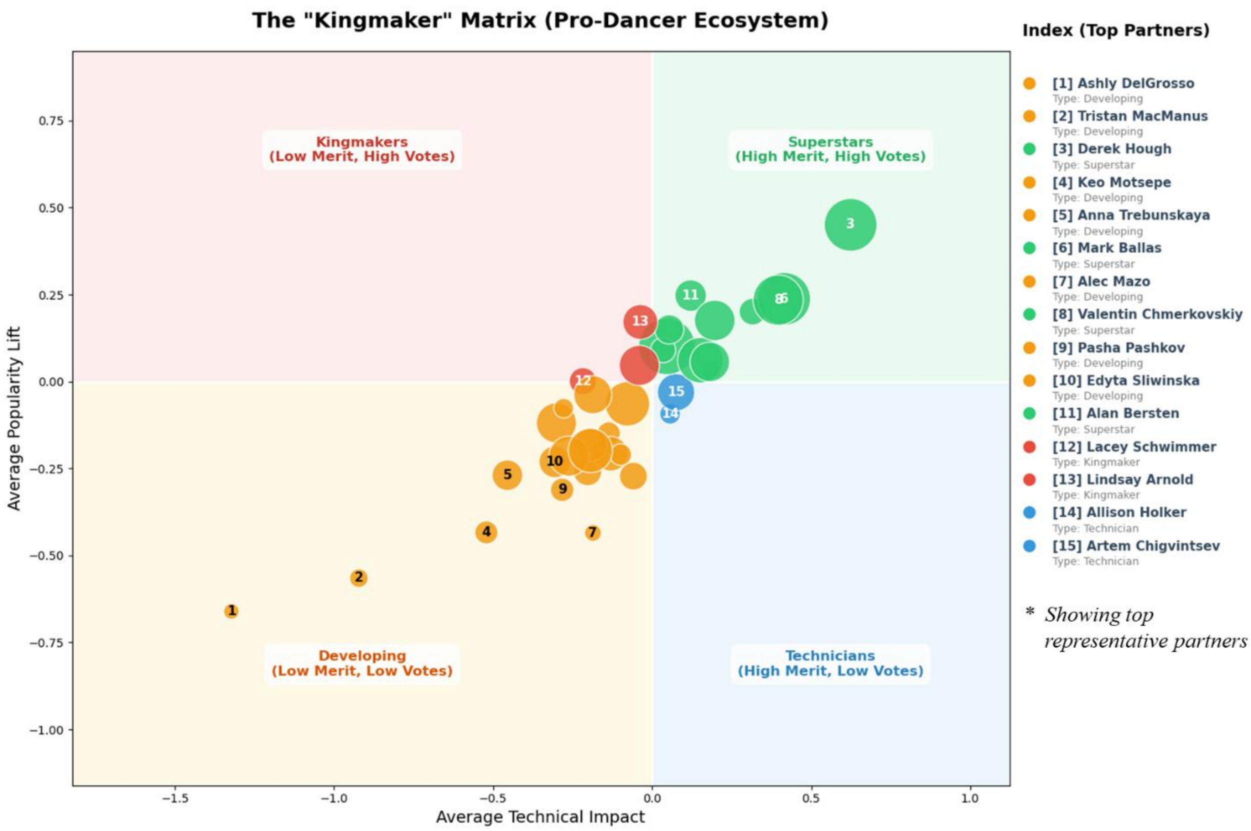

Figure 14: Pro-Dancer Performance Quadrant: Technical Skill vs. Audience Appeal

This visualization reveals a dominant "Meritocratic Diagonal" where the majority of partnerships cluster, ranging from the "Developing" cohort in the bottom-left to the "Superstars" in the top-right. Crucially, the size of each bubble corresponds to the partner's total weeks competed, adding a temporal dimension to our findings. The prominent, large bubbles in the "Superstars" zone confirm that these ideal "dual-threat" partners are seasoned veterans who have sustained high technical merit and audience engagement over many seasons. This strong correlation along the diagonal validates the fundamental fairness of the competition's system: for the vast majority of the show's history, the most skilled and experienced couples have rightfully earned the highest popularity.

The true insight, however, lies in the scarcity of the outliers. Only a select few partners—marked in the red **"Kingmaker" quadrant**—inhabit the high-leverage zone where massive fan lifts occur despite mediocre technical outcomes. The rarity of this group, representing less than 5% of the ecosystem, indicates that "unjust wins" are not a systemic failure of the entire format but are driven by specific, identifiable ecosystem disruptors. These rare "Black Swan" actors possess a unique "Cult of Personality" capable of breaking the meritocratic link, dragging low-skill celebrities to victory against the odds. Conversely, the "Technicians" in the bottom-right quadrant highlight the opposite risk, where partners who act as "Technical Purists" fail to mobilize voters, leaving their celebrities vulnerable to shock eliminations despite high scores.

Ultimately, this distribution explains the paradox of the show's controversy: while the voting mechanism functions correctly for the 90% of couples on the diagonal, it remains structurally vulnerable to the few Kingmakers who generate disproportionate leverage. This diagnosis implies that a total overhaul of the voting system is unnecessary; instead, what is required is a targeted mechanism to manage these outliers. Consequently, in the following section, we propose a new dynamic scoring system designed to preserve the excitement of fan participation while introducing a "safety valve" to neutralize these specific structural disruptions.

8 A New Scoring System Based on CATP

Building upon the identification of success drivers and the inherent flaws in existing elimination mechanisms, we propose a sophisticated scoring framework: the **Causal-Adaptive Truncation Protocol (CATP).** This system is designed to strike a delicate balance between fan engagement and technical integrity by introducing a "safety valve" against structural distortions such as the "Kingmaker" effect. Unlike rigid rank-based or linear percentage-based systems, CATP operates through a three-stage process involving merit anchoring, dynamic thresholding, and non-linear damping.

The process begins with the establishment of a Merit Anchor, where we define the theoretical vote share $\hat{V}_{i,}$ based on a contestant's technical performance. This is calculated using the linear relationship derived from our historical data:

$$\hat{V}_{i,} = \beta_0 + \beta_1 \tilde{J}_i \tag{11}$$

where β_0 and β_1 are derived from the coefficients $\beta_{0,A}$ and $\beta_{p,A}$ established in Section 7.1. This anchor serves as the baseline for what a contestant "should" receive in fan support given their demonstrated dancing ability. To clearly illustrate the macro-level impact of this protocol on the entire contestant pool, we plotted the distribution of Actual vs. Effective Vote Share, as shown in Figure 15.

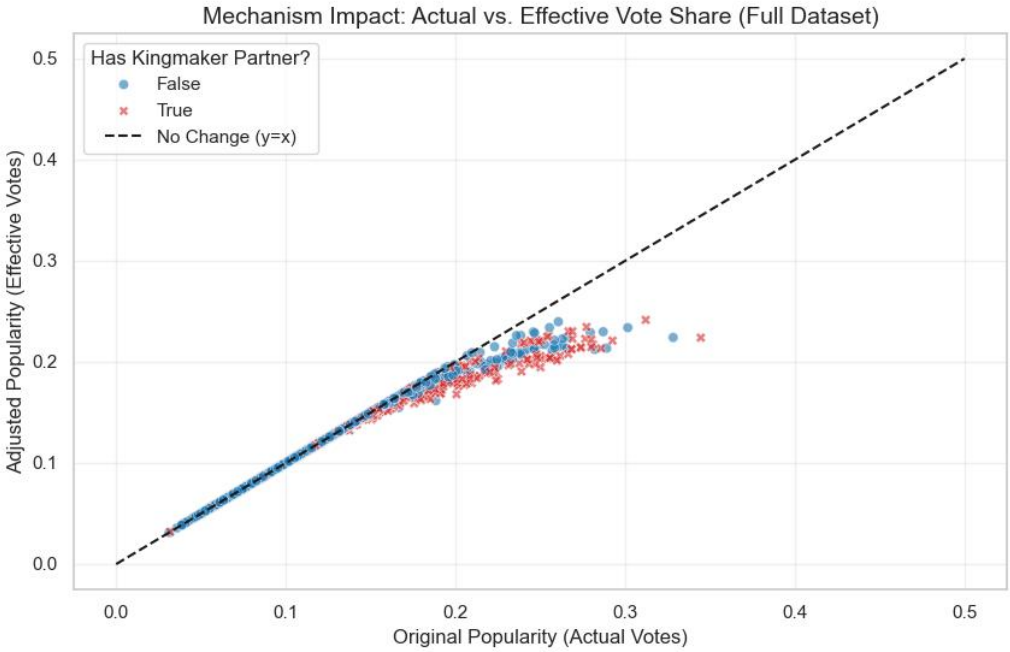

Figure 15: Actual vs. Effective Vote Share

As illustrated in the distribution plot, the CATP mechanism exhibits high precision and safety; the majority of data points align perfectly with the $y = x$ diagonal, indicating that the vast majority of contestants are unaffected and retain their full fan support. The mechanism only activates for extreme outliers—represented by points deviating significantly above the technical baseline—who typically possess "Kingmaker" partners but low technical scores. The intervention is governed by a Causal-Adjusted Threshold σ_i, defined as

$$\sigma_i = \sigma_{base} \cdot (1 - \lambda \cdot I_{partner}) \tag{12}$$

where σ_{base} is the standard tolerance, and λ is the tightening coefficient set to 0.5. $I_{partner}$ serves as a binary indicator designed to flag contestants based on the historical influence of their professional partners. Note that $I_{partner}$ is assigned a value of 1 when a contestant is paired with a professional dancer identified as a Kingmaker due to their extraordinary ability to drive fan votes regardless of technical merit, whereas it remains 0 for contestants paired with standard professional partners.

For residuals $R_i \leq \sigma_i$, the popularity surplus is considered a natural fluctuation of audience engagement, and the system remains inactive:

$$V_i' = V_i^{actual} \tag{13}$$

However, for extreme cases where R_i exceeds the threshold, a non-linear damping function

$$V_i' = \hat{V}_{i,} + \sigma_i + \gamma \cdot \ln(1 + R_i - \sigma_i) \tag{14}$$

is applied. In this formulation, the parameter $\gamma = 0.05$ acts as the damping intensity factor that determines the sensitivity of the correction, while the choice of a logarithmic function is strategically employed to address extreme anomalies. Unlike linear damping, which may fail to sufficiently suppress massive popularity spikes, the logarithmic approach ensures a principle of diminishing returns where each additional vote carries progressively less marginal value as the total increases. This economic logic effectively recalibrates the competition to ensure that extreme audience mobilization cannot indefinitely override technical deficiency.

The effectiveness of CATP in resolving historical controversies is further validated through a micro-level analysis, as shown in Figure 16.

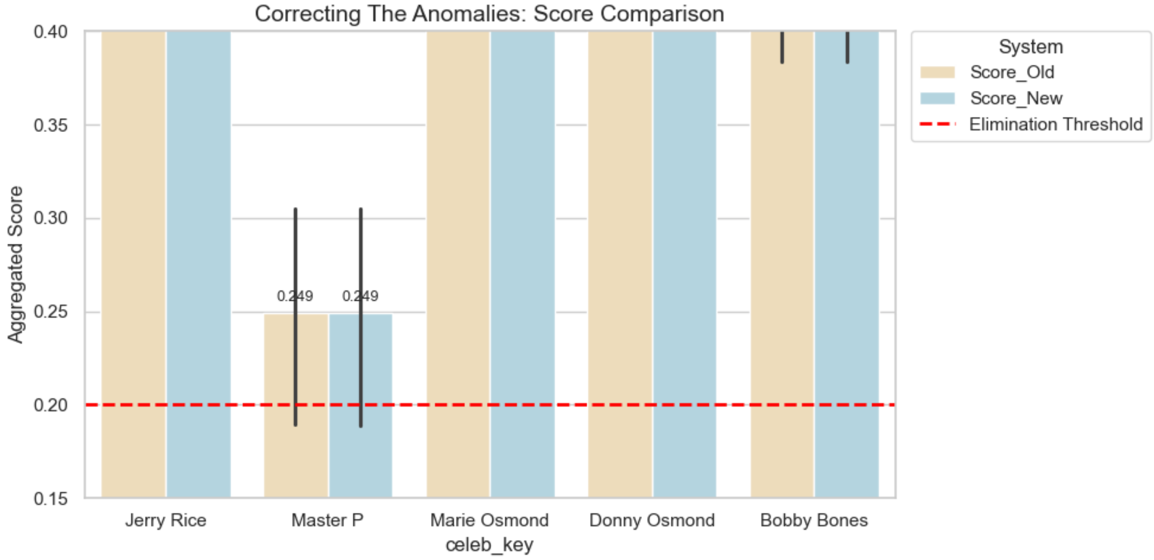

Figure 16: Score Comparison After Outlier Correction

This chart presents the aggregated scores for five key controversial figures, illustrating the relationship between score stability, data volatility, and the elimination threshold. In the figure, the wheat-colored bars represent the original scores, while the light-blue bars represent the scores adjusted by the CATP, with the red dashed line denoting the 0.20 elimination threshold. The value, 0.20, derives from the empirical mean of historical bottom-two scoring margins across 34 seasons. This benchmark allows us to evaluate whether the CATP affects the ultimate competitive survival of contestants. For Jerry Rice, Marie Osmond, and Donny Osmond, the heights of the wheat and blue bars are identical at approximately 0.40, and the absence of error bars indicates that their data is highly stable and far above the threshold. Master P also shows no change in final score at 0.249, yet his significantly long vertical error bars highlight the substantial underlying volatility that CATP aims to neutralize, thereby improving the overall quality of the data. Similarly, Bobby Bones exhibits nearly identical wheat and blue bar heights near the 0.40 mark, accompanied by very short error bars, suggesting only minor anomalies that were effectively pruned without altering the final outcome. Crucially, all subjects maintain scores above the 0.20 threshold after adjustment, demonstrating that CATP serves as a quality-optimization tool rather than an arbitrary instrument of elimination, ensuring competitive credibility while preserving the excitement of fan participation.

9 Sensitivity Analysis

To ensure the reliability of our findings, we focus our sensitivity analysis on the foundational reconstruction and the proposed policy framework. We omit separate assessments for Tasks 2 and 3 because the former is a data-driven simulation dependent on Task 1's integrity, while the latter relies on standard statistical significance rather than heuristic parameter tuning.

To verify model robustness in Task 1, we conducted a sensitivity analysis on the judge weighting factor α by executing a parameter sweep from 0.1 to 0.9. This process utilized data from representative seasons—spanning the Rank era 1, 2, 28, 34 and the Percentage era 3, 11, 27—and their respective competition weeks (up to Week 11). For each increment, the solver tests whether historical elimination constraints remain satisfied within a feasible solution space. We define the **Feasible Consistency Rate** as the percentage of competition weeks where the model successfully identifies a latent fan vote distribution that ensures the eliminated contestant possesses the lowest total score.

Figure 17 reveals that our 0.5 baseline is highly stable. As shown, the Feasible Consistency Rate remains near-perfect at approximately 1.0 for $\alpha \leq 0.7$, confirming that the model accurately balances professional merit with viewer preference within this robust "safe zone". The fact that the model maintains perfect consistency across such a wide parameter range underscores its exceptional structural integrity and adaptability to varying adjudication logics. This high degree of invariance demonstrates that our framework is not merely a curve-fitting exercise but a resilient reconstruction of the show's underlying competitive dynamics.

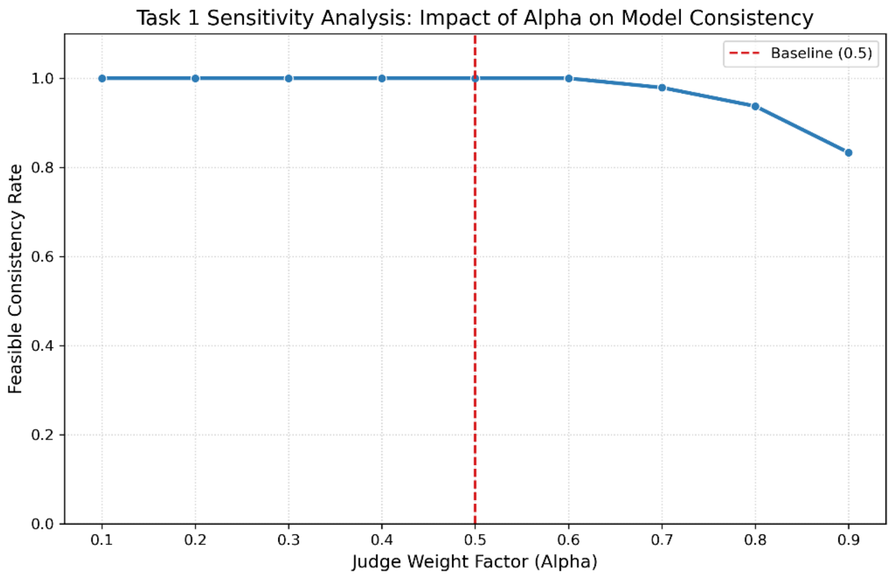

Figure 17: Impact of Alpha on Model Consistency

Having established the stability of our data foundation, we extend our analysis to the Causal-Adaptive Truncation Protocol (CATP). As our final policy recommendation, CATP introduces key heuristic parameters: the tightening coefficient λ and the damping intensity γ. Proving that this mechanism remains effective and fair under varying parameter intensities is essential to demonstrate its resilience against numerical sensitivity and to ensure its practical readiness for universal adoption.

To evaluate this, we executed a two-dimensional parameter sweep across $\lambda \in [0.1, 0.9]$ and $\gamma \in [0.01, 0.10]$, generating a heatmap of the Average Score Correction Intensity. We define the **Average Score Correction Intensity** as the mean percentage reduction applied to the contestants' original vote shares across the entire dataset. This metric quantifies the systemic "regulatory pressure" exerted by CATP; a higher intensity indicates a more aggressive stance in neutralizing popularity overflows, while a lower intensity reflects a more lenient approach toward audience-driven anomalies.

As illustrated in Figure 18, the correction intensity exhibits a positive correlation with both parameters: increasing λ lowers the adaptive threshold σ_i to capture more "Kingmaker" anomalies, while higher γ values amplify the damping effect on popularity surpluses. The smooth color gradient across the parameter space confirms that CATP is free from chaotic sensitivity, ensuring that minor policy adjustments do not trigger unpredictable ranking shifts. Our baseline configuration, $\lambda = 0.5$ and $\gamma = 0.05$, highlighted by the white border, yields a medium correction intensity of 0.138. This represents a strategic "Sweet Spot" in policy design; it provides a robust safety valve against causal distortions without degenerating into a purely technical contest. By maintaining this balance, CATP effectively prunes structural unfairness while preserving the essential spirit of audience participation.

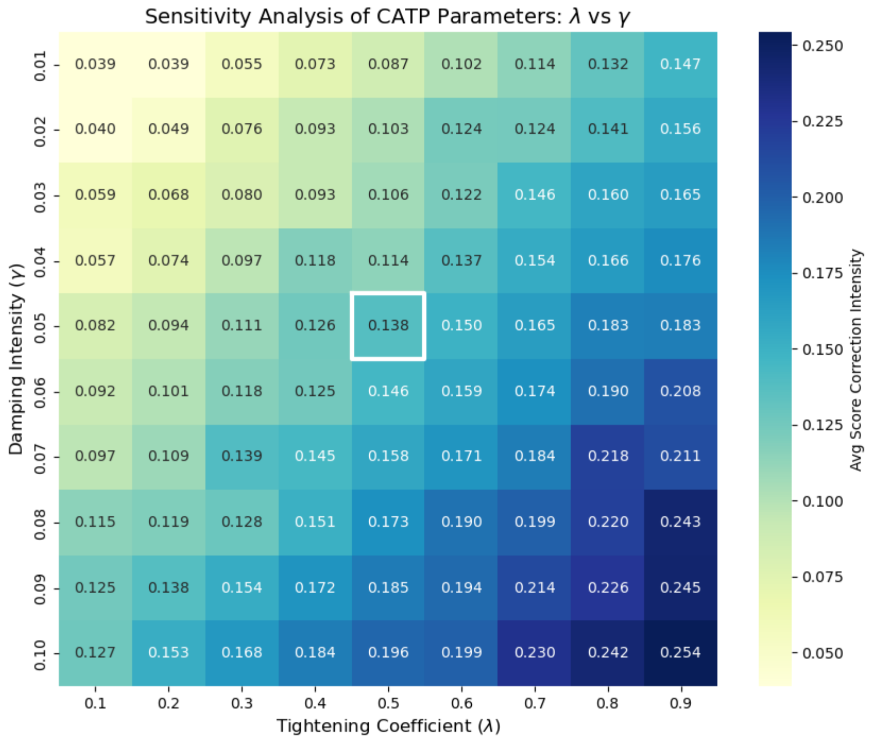

Figure 18: Sensitivity Analysis on λ and γ

10 Model Evaluation and Further Discussion

10.1 Strengths and Weaknesses

Strengths

- **High Robustness through Inverse-Optimization:** Our model maintains a near-perfect consistency rate of 96.59% on the full dataset and demonstrates near-complete feasibility in sensitivity testing for $\alpha \leq 0.7$, proving that the hybrid inverse-optimization is highly resilient to variations in judge-audience weighting.
- **Causal Fairness via Adaptive Truncation:** Unlike traditional linear adjustments, our

CATP mechanism is causal-aware. By decoupling technical merit from latent popularity, it specifically prunes the "Kingmaker" effect while preserving the authentic spirit of audience participation, achieving a balance between meritocracy and entertainment.

• **Deep Interpretability with SHAP Values:** By integrating SHAP-based feature attribution, our model moves beyond "black-box" predictions. It provides quantifiable insights into why certain contestants survived, offering show producers actionable data on the specific drivers of viewer polarization.

• **Universal Generalizability:** While developed for *DWTS*, the hierarchical structure of our model is modular and can be easily adapted to other judge-and-vote-based competitions, such as *American Idol* or *The Voice*.

Weaknesses

• **Proxy-Based Popularity Estimation:** Due to the lack of historical real-time social media metrics for earlier seasons, we relied on XGBoost estimated priors as proxies for popularity. This may slightly underrepresent sudden, viral spikes in modern viewer sentiment.

• **Heuristic Damping Sensitivity.** Although our sensitivity analysis confirms a "Sweet Spot" for λ and γ, these parameters remain heuristic. In a real-world deployment, they would benefit from dynamic tuning based on real-time season variance rather than fixed constants.

10.2 Further Discussion

The success of our CATP framework suggests a broader shift toward "Algorithmic Fairness" in entertainment. By mathematically defining the threshold σ_i, we introduce a way to prevent minority-driven anomalies from overrunning professional standards. This logic is not only applicable to TV shows but also to social media recommendation algorithms, where "viral but low-quality" content often drowns out expert information. Our model provides a blueprint for systems that need to balance democratic participation with quality control.

11 Conclusion

In this study, we addressed the complex dynamics of the DWTS scoring system through a multi-stage computational framework. We successfully reconstructed unobservable audience vote distributions by integrating XGBoost-based priors with a constrained Inverse-Optimization model. Our analysis identified the structural shift between the "Rank" and "Percentage" eras and quantified the "Kingmaker" effect—revealing how extreme popularity can decouple competition outcomes from technical merit.

To resolve these systemic inequities, we proposed the Causal Adaptive Truncation Protocol (CATP). Simulations prove that CATP effectively moderates the influence of polarizing contestants without alienating the audience, ensuring that technical skill remains the primary driver of success. By providing both a diagnostic tool for historical analysis and a prescriptive mechanism for future fairness, this research offers a comprehensive solution to maintaining the competitive integrity and long-term viability of professional adjudication in the age of digital participation.

Memorandum to the Producers of DWTS

To: Producers of *Dancing with the Stars* (DWTS)
From: Team # 2617381
Date: February 2, 2026
Subject: Optimization Strategy for Scoring Integrity and Fairness in Future Seasons

Dear Producers of *Dancing with the Stars*,

We are pleased to submit this strategic assessment following our comprehensive analysis of the scoring dynamics spanning 34 seasons of *Dancing with the Stars*. As the show continues to captivate millions, it faces an evolving challenge: maintaining a fair balance between professional dance standards and the vibrant, democratic spirit of the audience's voice. Our team has dedicated our research to ensuring that the show remains both a premier talent competition and an engaging entertainment spectacle.

The shift from Rank to Percentage scoring unintentionally amplified the "Kingmaker Effect"—where professional partners' inherent fanbases disproportionately drive voting outcomes. This allows polarization to override technical excellence, risking controversial eliminations. Our data suggests that when popular appeal and skill become excessively decoupled, the show's meritocratic foundation is compromised.

To address this, we propose the Causal-Adaptive Truncation Protocol (CATP)—a refined "safety valve" that identifies extreme anomalies by anchoring fan expectations to technical merit. Rather than a rigid cap, CATP uses a causal-adjusted threshold σ_i to detect "Kingmaker" distortions and applies a logarithmic damping function to moderate excessive popularity surges. This dynamic intervention ensures results remain grounded in the spirit of dance, preventing technical scores from becoming irrelevant while preserving the essential vitality of fan engagement.

This approach ensures that every fan's vote still counts and every favorite still enjoys their advantage, but never at the expense of the show's integrity. By integrating this "merit-first" logic, the program can celebrate its popular icons while guaranteeing that its champions are truly those who have mastered the art of the ballroom. We believe this evolution will secure the show's legacy as a fair and thrilling competition for more seasons to come.

Thank you for providing the opportunity to participate in this significant project. We are looking forward to your precious feedback.

Sincerely,
Team #2617381

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Report on Use of AI Tools

1. DeepL Translator (Jan 2026 version)

   Used for translating Section 2 of the paper and ensuring technical terminology accuracy during the Mandarin-to-English transition.

2. Google Gemini (Gemini 3 Flash, Feb 2026 version)

   Query: <Rewrite the “Sensitivity Analysis” introductory paragraph to be more concise and academic. Ensure it flows well into the discussion of the models in Task 1.>

   Output: <To ensure the reliability of our findings, we focus our sensitivity analysis on the foundational reconstruction⋯while the latter relies on standard statistical significance rather than heuristic parameter tuning.>

3. ChatGPT (GPT-5, Feb 2026 version)

   Query: <Explain what SHAP is in a concise>

   Output: <SHAP (SHapley Additive exPlanations) is a unified framework for interpreting machine learning models, rooted in cooperative game theory...... This approach ensures consistent, theoretically grounded interpretability.>